

Laboratory Guide 2: Entanglement & Multi-Qubit Systems

The "Spooky Action at a Distance"

Course: Quantum Computing Foundation

1 Introduction and Objectives

In Lab 1, we manipulated individual qubits. In this laboratory, we explore systems of multiple qubits and the phenomenon of **Entanglement**. As stated in Lesson QC05, the real power of quantum computation appears when we start combining qubits together.

Prerequisites:

- Lesson QC05: *Multiple Qubits*
- Lab 1: *Introduction to QISKIT*

Learning Objectives:

1. Understand the Controlled-NOT (CNOT) gate logic.
2. Create a Bell State (Maximally Entangled State) using the Composer.
3. Visualize correlations between qubits on the Q-Sphere.
4. Verify entanglement statistics using Qiskit code.
5. Extend entanglement to 3 qubits (GHZ State) and discuss scalability.

Important: Qiskit Bit Ordering

Reading Order in Qiskit

Qiskit prints measurement results as bit strings in "Little Endian" order: the rightmost bit is qubit 0 (q_0).

$$\text{Result } 10 \implies q_1 = 1, \quad q_0 = 0$$

The kets in this guide are written the other way round, $|q_0q_1\rangle$ (first symbol = qubit 0): the ket $|10\rangle$ means $q_0 = 1, q_1 = 0$, which Qiskit prints as 01. Always pay attention to which qubit is the *Control* and which is the *Target*.

2 Part A: Logic & Entanglement (IBM Quantum Composer)

Platform: IBM Quantum Composer

2.1 Exercise 1: The CNOT Gate (Conditional Logic)

Context: Lesson **QC05** introduces the CNOT gate. It acts on two qubits: a *Control* and a *Target*. It flips the Target qubit ONLY IF the Control qubit is in state $|1\rangle$.

Student Activity

Setup: A new Composer circuit has four qubits. Reduce it to two: in the OpenQASM editor change `qubit[4] q;` to `qubit[2] q;` and `bit[4] c;` to `bit[2] c;` (or use *Edit* → *Manage registers*). With four qubits every label has four digits (11 is shown as 0011).

Action:

1. Use an **X** gate on q_0 (Control) to set it to $|1\rangle$. Leave q_1 (Target) at $|0\rangle$.
2. Drag a **CNOT** gate (symbol $\oplus$ connected to a dot). Make sure the **dot (control)** is on q_0 and the **plus (target)** is on q_1 .
3. Observe the Output State.

Expected Effect: The state becomes $|11\rangle$ (Probability 100% on "11"). The target flipped because the control was 1.

2.2 Exercise 2: Creating the Bell State (Entanglement)

Context: Entanglement is a state where two particles become inextricably linked. The most famous example is the **Bell State** $|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$.

Student Activity

Action:

1. Clear the circuit.
2. Apply an **H** gate to q_0 . (State is now superposition of 0 and 1).
3. Apply a **CNOT** gate (Control q_0 , Target q_1).
4. Check the **Probabilities**.

Expected Effect: You should see two bars:

- $\approx 50\%$ Probability for $|00\rangle$
- $\approx 50\%$ Probability for $|11\rangle$
- **0% Probability** for $|01\rangle$ or $|10\rangle$.

Correlation is not yet entanglement. Two coins put in a box, both heads or both tails, 50/50, give the same 00/11 histogram. What no pair of separate qubit states can do is produce $|\Phi^+\rangle$. Try $(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$. To match $|\Phi^+\rangle$ we need $ad = 0$ and $bc = 0$ (no 01, no 10), but also $ac \neq 0$ and $bd \neq 0$ (00 and 11 present). If $ad = 0$, then $a = 0$ or $d = 0$, and either one makes ac or bd zero: a contradiction, so the state cannot be factored. The difference is measurable: apply H to *both* qubits just before measuring. $|\Phi^+\rangle$ still

gives only 00 and 11, because $(H \otimes H)|\Phi^+\rangle = |\Phi^+\rangle$, while the box of coins would give all four outcomes, 25% each. Try it in the Composer.

IBM Composer View (Q-Sphere)

Q-Sphere Visualization

Look at the Q-Sphere: two dots, $|00\rangle$ and $|11\rangle$, same size, same colour. The dots tell you which outcomes are possible, and with which sign. They do **not** tell you on their own whether the state is entangled: try H on q_1 only, and you also get two dots, yet each qubit then has a state of its own. What makes $|\Phi^+\rangle$ special is *which* pair survives: the two qubits always agree. As the paragraph *Correlation is not yet entanglement* above shows, no choice of separate single-qubit states can produce that.

Deep Dive: Why does this matter?

Why is Entanglement so Important?

Entanglement is not just a weird phenomenon; it is the fundamental resource for two game-changing applications.

1. Computational Power (The "Speedup")

As we will see in **Lesson QC14** (Algorithms), entanglement allows a quantum computer to process a vast state space simultaneously. Here a brief description to some of those algorithms, that extensively makes use of the entanglement effect:

- **Shor's Algorithm:** Uses entanglement to crack RSA encryption exponentially faster than classical computers.
- **Grover's Algorithm:** Uses entanglement to search through unsorted databases with quadratic speedup.
- *Hardware Context:* Chips like **Google Willow** (December 2024, 105 qubits) use entangled qubits to perform tasks impossible for classical supercomputers.

2. Unhackable Security (Quantum Internet)

Entanglement is one route to **Quantum Key Distribution (QKD)**: entanglement-based protocols use shared pairs, while the first QKD protocol, BB84 (1984), sends single photons and needs no entanglement (Lab 6).

- If two parties share entangled photons, any attempt by a hacker to "eavesdrop" (measure the line) destroys the entanglement instantly.
- The disturbance shows up as errors: when the two parties measure in randomly chosen bases (as real protocols do, Lab 6), an intercepted round they keep gives a wrong bit in one case out of four, so by comparing a random sample of their bits the two parties detect the intruder with a probability that approaches 1 as the sample grows.
- *Hardware Context:* Companies such as **IonQ** are building the nodes for these quantum networks.

3 Part B: Coding Entanglement (Qiskit)

3.1 Exercise 3: The "Hello World" of Quantum Code

We will now write the Python code to generate the Bell State. This is the standard test to verify a quantum computer works.

Coding Task: Bell State

```
1 from qiskit import QuantumCircuit
2 from qiskit_aer import AerSimulator
3 from qiskit.visualization import plot_histogram
4
5 # 1. Initialize Circuit with 2 Qubits
6 qc = QuantumCircuit(2, 2)
7
8 # 2. H on q0, then CNOT(0->1)
9 qc.h(0)
10 qc.cx(0, 1)
11
12 # 3. Measure both
13 qc.measure([0, 1], [0, 1])
14
15 # 4. Execute
16 backend = AerSimulator()
17 job = backend.run(qc, shots=1000)
18 print(f"Bell Results: {job.result().get_counts()}")
```

3.2 Exercise 4: Scaling Up - The GHZ State (3 Qubits)

Context: State-of-the-art quantum processors (e.g., **Google Willow**, **IonQ Tempo**) rely on creating entanglement across many qubits. The simplest example of *multipartite entanglement* is the **GHZ State**:

$$|GHZ\rangle = \frac{|000\rangle + |111\rangle}{\sqrt{2}}$$

Concept Preview: What is Decoherence?

You will study this formally in **Lesson QC06**, but for now, think of **Decoherence** as "environmental noise".

- Just as a spinning top eventually falls due to friction, quantum states lose their superposition due to interaction with the outside world (heat, radiation).
- **The Rule:** The more qubits you entangle, the faster the system decoheres. This is why a 3-qubit GHZ state is much more fragile than a 2-qubit Bell state.

Coding Task: The 3-Qubit Entangler

```
1 # 1. Initialize 3 qubits
2 qc3 = QuantumCircuit(3, 3)
3
4 # 2. Entangle q0 and q1 (Bell State)
5 qc3.h(0)
6 qc3.cx(0, 1)
7
8 # 3. Entangle q2 (Chain reaction: q1 controls q2)
9 qc3.cx(1, 2)
10
11 # 4. Measure
12 qc3.measure([0, 1, 2], [0, 1, 2])
13
14 # 5. Execute
15 job3 = AerSimulator().run(qc3, shots=1000)
16 print(f"GHZ Results: {job3.result().get_counts()}")
```

Expected Analysis: Ideal results are '000' and '111'. In a real system, you would see errors (e.g., '001') appearing more frequently here than in the previous exercise, exactly due to the decoherence effect described above.

4 Part C: Assessment (Quiz)

Q1. The Logic of CNOT

If the control qubit is in state $|0\rangle$ and the target qubit is in state $|1\rangle$, what does the CNOT gate do?

- A) Flips the target to $|0\rangle$.
- B) Flips the control to $|1\rangle$.
- C) Nothing. The state remains $|01\rangle$ (Control is 0, so Target is untouched).

Q2. Instantaneous Correlation

You have created the Bell State $\frac{|00\rangle + |11\rangle}{\sqrt{2}}$. You measure the first qubit and get the result "1". What is the state of the second qubit immediately after?

- A) Still 50% "0" and 50% "1".
- B) Definitely "1". The wavefunction has collapsed to $|11\rangle$.
- C) Definitely "0".

Q3. The Fragility of Large Systems

In Exercise 4, we mentioned that "Decoherence" acts like noise. Why is the 3-qubit GHZ state harder to maintain than a single qubit?

- A) Because Qiskit cannot handle 3 qubits.
- B) Because having more entangled particles increases the interaction surface with the environment, making the system lose information faster.
- C) Because 3 is an odd number.

Solutions: Q1: C; Q2: B; Q3: B (More qubits = Faster decoherence).